



SMART INDIA
HACKATHON
2026

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VEDAVANGUARD

Multilingual RAG-Based Ayurveda IPR Assistant

- **Problem Statement ID** – SIH26045
- **Theme** – Healthcare / Traditional Knowledge
- **PS Category** – Software
- **Solution** – Source-grounded IPR guidance across English, Hindi, Sanskrit, and regional languages



VEDAVANGUARD

A multilingual, source-grounded IPR assistant for Ayurveda.

VedaVanguard translates a user's question into a jurisdiction-aware answer with citations, confidence, and safe abstention across English, Hindi, Sanskrit, and regional languages.

Ask

Describe the formulation or question in English, Hindi, Sanskrit, or a regional language.

Classify

Identify the formulation type and choose the India or international jurisdiction.

Retrieve

Connect the question to the right legal, traditional-knowledge, and prior-art sources.

Act

Return a cited next step with confidence, or abstain safely when evidence is not strong enough.

From a multilingual question to a defensible next step

TECHNICAL APPROACH

01

Input

Multilingual query plus India / international jurisdiction toggle.

02

Classify

Pre-RAG formulation classifier: Classical, Proprietary, New Drug, Phytopharma, or Cosmetic.

03

Retrieve

Hybrid dense + BM25 retrieval over a curated legal corpus in ChromaDB with jurisdiction metadata filtering.

04

Guardrail

Citations, confidence score, safe abstention, disclaimer, and DPDP audit log.

05

Extend

Compliance modules: ABS helper, TKDL prior-art mapper, and knowledge-graph multi-hop reasoning.

Feasible by design, shippable on a student-team timeline

FEASIBILITY AND VIABILITY

Mature stack

FastAPI, ChromaDB, MiniLM-L12-v2, Gemini/Groq, and Bhashini APIs.

Accessible corpus

Patent Act, Biodiversity Act, GI Registry, API, TKDL, WIPO GRATK 2024, and Nagoya materials.

Language ready

Bhashini supports translation, ASR, and TTS APIs across 22 Indian languages.

Delivery

6–8 weeks, 2–3 developers, no GPU, serverless-friendly, about \$30/month.

Market pull

50K+ AYUSH/MSME users plus SaaS, API, white-label, and grant pathways.

Buildable → Deployable → Fundable

Protect knowledge. Reduce risk. Expand access.

IMPACT AND BENEFITS

01

Core Idea

Make traditional knowledge legible to modern IP workflows.

02

Key Benefits

Faster, safer, more inclusive guidance.

03

Supporting

Data

Traceable sources, confidence, and auditability.

04

Action Steps

Turn insight into a compliance workflow.

RESEARCH AND REFERENCES

Built on public law, traditional texts, and traceable retrieval

Knowledge and prior art

- Traditional Knowledge Digital Library — CSIR and Ministry of AYUSH
- Charaka Samhita, Sushruta Samhita, Ashtanga Hridaya; CCRAS pharmacopoeia and formulary
- Geographical Indications Registry

Law and policy

- India Patent Act, 1970, Section 3(p)
- Biological Diversity Act, 2002; Access and Benefit Sharing Regulations, 2014
- Nagoya Protocol; TRIPS Article 27.3(b)

Retrieval and language

- WIPO genetic resources and traditional knowledge work; WIPO GRATK 2024
- Multilingual MiniLM; ChromaDB; Groq Cloud
- Bhashini translation, ASR, and TTS APIs